Reflected SDE and BSDE

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Preface

This is the reading note on reflected stochastic differential equations. Reflected SDEs are introduced in Chapters 1, 2, and 3, which are mainly from Andrey Pilipenko's note *An Introduction to Stochastic Differential Equations with Reflection*. Reflected BSDEs are introduced in Chapter 4, which is mainly from Chapter 6 of Jianfeng Zhang's book *Backward Stochastic Differential Equations: From Linear to Fully Nonlinear Theory*.

Contents

Chapter 1 One-Dimensional Reflected SDEs

1.1 The Skorokhod problem on the half-line

Consider a continuous driving path on the nonnegative half-line. Whenever the unconstrained path attempts to cross the boundary at zero, a minimal nondecreasing correction pushes it back into the state space.

Definition 1.1 (Skorokhod problem on $[0, \infty)$)**.** Let $f \in C([0, T]; \mathbb{R})$ with $f(0) \geq 0$. A pair $(g, \ell) \in C([0, T]; \mathbb{R})^2$ solves the Skorokhod problem for f if

(i) $g(t) \geq 0$ for every $t \in [0, T]$;

(ii) $g(t) = f(t) + \ell(t)$ for every $t \in [0, T]$;

(iii) $\ell(0) = 0$ and ℓ is nondecreasing;

(iv)

$$\int_0^T \mathbf{1}_{\{g(s)>0\}} \, \mathrm{d}\ell(s) = 0.$$

The last condition says that the regulator may increase only while the reflected path is at the boundary.

Theorem 1.2 (Explicit solution)**.** *For every $f \in C([0, T]; \mathbb{R})$ with $f(0) \geq 0$, the Skorokhod problem has a unique solution. It is given by*

$$\ell(t) = \sup_{0 \leq s \leq t} [-f(s)]^+ = -\min_{0 \leq s \leq t} \big(f(s) \wedge 0\big), \tag{1.1}$$

$$g(t) = f(t) + \ell(t) = f(t) - \min_{0 \leq s \leq t} \big(f(s) \wedge 0\big). \tag{1.2}$$

Proof. Define ℓ by (1.1) and set $g = f + \ell$. The running supremum of the continuous function f^- is continuous and nondecreasing, and $\ell(0) = 0$ because $f(0) \geq 0$. Moreover, $\ell(t) \geq [-f(t)]^+$, so $g(t) \geq 0$.

The Stieltjes measure $\mathrm{d}\ell$ is supported on the set of times at which f^- attains a new running maximum. At each such time, $\ell(t) = -f(t)$ and hence $g(t) = 0$. Therefore $\int_0^T \mathbf{1}_{\{g>0\}} \, \mathrm{d}\ell = 0$, proving existence.

For uniqueness, let $(\widetilde{g}, \widetilde{\ell})$ be any solution. Since $\widetilde{g} = f + \widetilde{\ell} \geq 0$,

$$\widetilde{\ell}(t) \geq \sup_{0 \leq s \leq t} [-f(s)]^+ = \ell(t).$$

If strict inequality held at some time, continuity would give a first level at which $\widetilde{\ell}$ rises above ℓ. At that increase time the support condition forces $\widetilde{g} = 0$, so $\widetilde{\ell} = -f \leq \sup_{s \leq t}[-f(s)]^+ = \ell$, a contradiction. Thus $\widetilde{\ell} = \ell$ and consequently $\widetilde{g} = g$. □

Definition 1.3 (Skorokhod map)**.** For $f \in C([0,T];\mathbb{R})$ define

$$(\Lambda f)(t) := \sup_{0 \leq s \leq t} [-f(s)]^+, \qquad (\Gamma f)(t) := f(t) + (\Lambda f)(t).$$

The map Γ is the one-dimensional Skorokhod map and Λf is its regulator. The formulas make sense for every continuous f. When $f(0) < 0$, $\Lambda f(0) > 0$, so this extended map no longer uses the normalization $\ell(0) = 0$ from the preceding definition.

For $h \in C([0,T];\mathbb{R})$ write

$$\|h\|_t := \sup_{0 \leq s \leq t} |h(s)|, \qquad \omega_h(\delta; t) := \sup_{\substack{r,s \in [0,t] \\ |r-s| \leq \delta}} |h(r) - h(s)|.$$

Proposition 1.4 (Stability of the one-dimensional map)**.** *For $f_1, f_2 \in C([0,T];\mathbb{R})$ and $t \in [0,T]$,*

$$\|\Lambda f_1 - \Lambda f_2\|_t \leq \|f_1 - f_2\|_t, \tag{1.3}$$

$$\|\Gamma f_1 - \Gamma f_2\|_t \leq 2\,\|f_1 - f_2\|_t. \tag{1.4}$$

In addition,

$$\|\Lambda f\|_t \leq \|f\|_t, \qquad \|\Gamma f\|_t \leq 2\,\|f\|_t, \tag{1.5}$$

$$\omega_{\Lambda f}(\delta; t) \leq \omega_f(\delta; t), \qquad \omega_{\Gamma f}(\delta; t) \leq \omega_f(\delta; t). \tag{1.6}$$

Consequently, both Λ and Γ are nonanticipative and continuous for the uniform topology on compact time intervals.

Proof. The map $x \mapsto x^+$ is 1-Lipschitz, and the running-supremum operator is 1-Lipschitz in the supremum norm. This proves (1.3); adding the driving paths gives (1.4). The bounds (1.5) follow by comparing with the zero path.

Let $0 \leq s < t \leq T$. If $\Lambda f(t) = \Lambda f(s)$, then both increments in (1.6) are immediate. Otherwise choose $u \in (s,t]$ such that $\Lambda f(t) = [-f(u)]^+$. Since $\Lambda f(s) \geq [-f(s)]^+$,

$$0 \leq \Lambda f(t) - \Lambda f(s) \leq |f(u) - f(s)| \leq \omega_f(t-s; T).$$

Furthermore, both $\Gamma f(s)$ and $\Gamma f(t)$ lie between zero and the oscillation of f over $[s,t]$ relative to the new minimum at u; hence $|\Gamma f(t) - \Gamma f(s)| \leq \omega_f(t-s; T)$. Taking suprema proves (1.6). □

Remark 1.5. The same problem can be formulated for càdlàg paths. With jumps, one must specify how the regulator reacts to an overshoot of the boundary; the continuous theory considered here avoids that additional convention.

1.2 One-dimensional reflected stochastic differential equations

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions, and let B be a one-dimensional $(\mathcal{F}_t)$-Brownian motion. Let $a, b : [0,\infty) \times [0,\infty) \to \mathbb{R}$ be measurable, and let $\xi_0 \geq 0$ be $\mathcal{F}_0$-measurable.

Definition 1.6 (Strong solution)**.** A pair of continuous adapted processes (X, L) is a strong solution of the reflected SDE

$$X_t = \xi_0 + \int_0^t a(s, X_s)\, \mathrm{d}s + \int_0^t b(s, X_s)\, \mathrm{d}B_s + L_t \tag{1.7}$$

if $X_t \geq 0$, $L_0 = 0$, L is nondecreasing,

$$\int_0^t \mathbf{1}_{\{X_s>0\}}\, \mathrm{d}L_s = 0 \qquad \text{for every } t \geq 0,$$

and all terms in (1.7) are well defined.

If

$$Y_t := \xi_0 + \int_0^t a(s, X_s)\, \mathrm{d}s + \int_0^t b(s, X_s)\, \mathrm{d}B_s,$$

then pathwise uniqueness for the Skorokhod problem gives

$$X = \Gamma Y, \qquad L = \Lambda Y.$$

Because the maps are nonanticipative, applying them to a continuous adapted process again produces continuous adapted processes.

Theorem 1.7 (Existence and pathwise uniqueness)**.** *Assume that, for constants $C, \lambda > 0$ and all $t \geq 0$ and $x, y \geq 0$,*

$$\begin{aligned} |a(t,x) - a(t,y)| + |b(t,x) - b(t,y)| &\leq \lambda |x-y|, \\ |a(t,x)| + |b(t,x)| &\leq C(1+x). \end{aligned}$$

If $\xi_0 \in L^2(\mathcal{F}_0)$ and $\xi_0 \geq 0$, then (1.7) has a unique strong solution on every finite time interval.

Proof. For an adapted continuous process U, define

$$(\Phi U)_t := \Gamma\left(\xi_0 + \int_0^\cdot a(s, U_s)\, \mathrm{d}s + \int_0^\cdot b(s, U_s)\, \mathrm{d}B_s\right)(t).$$

The Lipschitz estimate for Γ, the Burkholder–Davis–Gundy inequality, and the coefficient Lipschitz bound imply, on $[0, h]$,

$$\mathbb{E} \sup_{t\leq h} |(\Phi U)_t - (\Phi V)_t|^2 \leq C_\lambda h\, \mathbb{E} \sup_{t\leq h} |U_t - V_t|^2.$$

For sufficiently small h, Φ is a contraction on the usual space of continuous adapted processes with finite L^2 supremum norm. The linear growth bound shows that Φ maps this space into itself. Banach's fixed point theorem gives a solution on $[0, h]$. Restarting the Skorokhod problem from the current nonnegative state extends the solution over any finite horizon. The same estimate, followed by Gronwall's lemma, proves pathwise uniqueness. $\square$

Theorem 1.8 (Moment and stability estimates). *Let the preceding assumptions hold, let $p \geq 2$, and suppose $\xi_0 \in L^p$. For each finite T there is a constant $C_{p,T}$, depending only on p, T, C, and λ, such that*

$$\mathbb{E}\left[\sup_{0\leq s\leq T} |X_s|^p + L_T^p\right] \leq C_{p,T}\left(1 + \mathbb{E}|\xi_0|^p\right). \tag{1.8}$$

If (X^x, L^x) and (X^y, L^y) are driven by the same Brownian motion and start from deterministic $x, y \geq 0$, then

$$\mathbb{E}\left[\sup_{0\leq s\leq T} |X_s^x - X_s^y|^p + \sup_{0\leq s\leq T} |L_s^x - L_s^y|^p\right] \leq C_{p,T}|x-y|^p. \tag{1.9}$$

Finally, for $0 \leq t < t+h \leq T$ with $h \leq 1$,

$$\mathbb{E}\left[\sup_{t\leq s\leq t+h} |X_s - X_t|^p + |L_{t+h} - L_t|^p\right] \leq C_{p,T}\left(1 + \mathbb{E}|\xi_0|^p\right) h^{p/2}. \tag{1.10}$$

Proof. Write the solution as $X = \Gamma Y$ and $L = \Lambda Y$. The map estimates reduce all three claims to the corresponding drift and stochastic-integral bounds for Y. Hölder's inequality controls the drift, the Burkholder–Davis–Gundy inequality controls the martingale part, and Gronwall's lemma yields (1.8) and (1.9). For (1.10), restart the Skorokhod problem at time t and apply the same inequalities on the interval $[t, t+h]$; the drift contributes order h^p, which is bounded by order $h^{p/2}$ when $h \leq 1$. $\square$

1.2.1 The Markov property

For $s \leq t$ and $x \geq 0$, let $X^{s,x}$ denote the unique solution started from x at time s and driven by the Brownian increments $(B_r - B_s)_{r\geq s}$. Define

$$P_{s,t}(x, A) := \mathbb{P}\left(X_t^{s,x} \in A\right), \qquad A \in \mathcal{B}([0,\infty)).$$

Theorem 1.9 (Time-inhomogeneous Markov property). *Under the Lipschitz and growth assumptions, the reflected diffusion satisfies*

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = P_{s,t}(X_s, A) \qquad \mathbb{P}\text{-a.s.}, \qquad 0 \leq s \leq t.$$

Thus $(P_{s,t})_{s\leq t}$ is the transition family of X. If the coefficients are time homogeneous, this family reduces to a Markov semigroup.

Proof. The future Brownian increments are independent of $\mathcal{F}_s$. Pathwise uniqueness and the nonanticipativity of the Skorokhod map give the flow identity

$$X_t = X_t^{s,X_s} \qquad \mathbb{P}\text{-a.s.}$$

Conditioning a bounded Borel function of the right-hand side on $\mathcal{F}_s$ therefore yields the displayed formula. $\square$

1.3 Connection with local time

We use the right local-time convention. For a continuous semimartingale Z, its right local time at a is the jointly measurable version satisfying

$$L_t^{a,+}(Z) = \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon} \int_0^t \mathbf{1}_{[a,a+\varepsilon)}(Z_s)\ \mathrm{d}\langle Z\rangle_s, \tag{1.11}$$

whenever the limit is interpreted in the standard local-time sense. It obeys Tanaka's formula

$$(Z_t - a)^+ = (Z_0 - a)^+ + \int_0^t \mathbf{1}_{\{Z_s > a\}}\ \mathrm{d}Z_s + \frac{1}{2} L_t^{a,+}(Z), \tag{1.12}$$

and the occupation-density formula

$$\int_0^t q(Z_s)\ \mathrm{d}\langle Z\rangle_s = \int_{\mathbb{R}} q(a) L_t^{a,+}(Z)\ \mathrm{d}a$$

for nonnegative Borel q; changing the local-time version on a null set of levels does not alter the last integral.

Theorem 1.10 (The regulator is boundary local time)**.** *Let (X, L) solve (1.7). Assume $b(t,0) \neq 0$ for Lebesgue-almost every t. Then, simultaneously for all $t \geq 0$,*

$$L_t = \frac{1}{2} L_t^{0,+}(X) \qquad \mathbb{P}\text{-a.s.} \tag{1.13}$$

Equivalently, L is the symmetric local time of the nonnegative process X at the boundary.

Proof. Since $X = X^+$, Tanaka's formula and the support condition for $\mathrm{d}L$ give

$$\begin{aligned} \frac{1}{2} L_t^{0,+}(X) &= X_t - X_0 - \int_0^t \mathbf{1}_{\{X_s > 0\}}\ \mathrm{d}X_s \\ &= L_t + \int_0^t \mathbf{1}_{\{X_s = 0\}} a(s,0)\ \mathrm{d}s + \int_0^t \mathbf{1}_{\{X_s = 0\}} b(s,0)\ \mathrm{d}B_s. \end{aligned}$$

The occupation-density formula applied to the singleton $\{0\}$ yields

$$\int_0^t \mathbf{1}_{\{X_s = 0\}} b(s,0)^2\ \mathrm{d}s = 0.$$

Because $b(s,0) \neq 0$ almost everywhere, X spends zero Lebesgue time at zero. The drift term above vanishes, and the stochastic integral has zero quadratic variation and hence vanishes. This proves (1.13). $\square$

Example 1.11 (Lévy's identity)**.** Let W be Brownian motion and let $L^0(W)$ denote its symmetric local time at zero. Tanaka's formula gives

$$|W_t| = \widetilde{B}_t + L^0_t(W), \qquad \widetilde{B}_t := \int_0^t \operatorname{sgn}(W_s)\ \mathrm{d}W_s,$$

where $\widetilde{B}$ is Brownian motion by Lévy's characterization. The pair $(|W|, L^0(W))$ solves the Skorokhod problem for $\widetilde{B}$, and therefore

$$L^0_t(W) = -\inf_{0\le s\le t} \widetilde{B}_s, \qquad |W_t| = \widetilde{B}_t - \inf_{0\le s\le t} \widetilde{B}_s.$$

Consequently,

$$\big(|W_t|, L^0_t(W)\big)_{t\ge 0} \overset{d}{=} \left(B_t - \inf_{0\le s\le t} B_s, -\inf_{0\le s\le t} B_s\right)_{t\ge 0}.$$

Chapter 2 Multidimensional Reflected Stochastic Differential Equations

2.1 Normal reflection in a half-space

Let

$$\mathbb{R}^d_+ := \mathbb{R}^{d-1} \times [0, \infty), \qquad e_d := (0, \dots, 0, 1).$$

Definition 2.1 (Half-space Skorokhod problem). Let $f = (f^1, \dots, f^d) \in C([0, \infty); \mathbb{R}^d)$ with $f(0) \in \mathbb{R}^d_+$. A pair $(g, \ell) \in C([0, \infty); \mathbb{R}^d) \times C([0, \infty); \mathbb{R})$ solves the normal-reflection Skorokhod problem for f if

(i) $g(t) \in \mathbb{R}^d_+$ for every $t \geq 0$;

(ii) $g(t) = f(t) + e_d \ell(t)$;

(iii) $\ell(0) = 0$, ℓ is nondecreasing, and for every $t \geq 0$,

$$\int_0^t \mathbf{1}_{\{g(s) \notin \partial \mathbb{R}^d_+\}} \, \mathrm{d}\ell(s) = 0.$$

Proposition 2.2 (Explicit half-space map). *The half-space problem has the unique solution*

$$\begin{aligned} g^i(t) &= f^i(t), && i = 1, \dots, d-1, \\ \ell(t) &= -\min_{0 \leq s \leq t} \big(f^d(s) \wedge 0\big), \\ g^d(t) &= f^d(t) + \ell(t). \end{aligned}$$

Consequently, on every finite interval $[0, T]$, the reflected path and the regulator satisfy the same Lipschitz and modulus estimates as the one-dimensional Skorokhod map in the last coordinate.

Proof. Only the last coordinate is constrained. Apply the one-dimensional theorem to f^d and leave the first $d-1$ coordinates unchanged. □

Let B be an m-dimensional Brownian motion and let $a : [0, T] \times \mathbb{R}^d_+ \to \mathbb{R}^d$ and $\sigma : [0, T] \times \mathbb{R}^d_+ \to \mathbb{R}^{d \times m}$ be measurable. A reflected SDE in the half-space has the form

$$X_t = X_0 + \int_0^t a(s, X_s) \, \mathrm{d}s + \int_0^t \sigma(s, X_s) \, \mathrm{d}B_s + e_d L_t. \tag{2.1}$$

Here $X_t \in \mathbb{R}^d_+$, L is continuous and nondecreasing with $L_0 = 0$, and $\mathrm{d}L$ is supported on $\{t : X_t \in \partial\mathbb{R}^d_+\}$. Global Lipschitz and linear-growth assumptions on a and σ give existence and pathwise uniqueness by the same fixed-point argument as in one dimension.

2.2 Geometry of normal reflection in a domain

Let $D \subset \mathbb{R}^d$ be a nonempty open connected set. The reflection direction will point into $\overline{D}$.

Definition 2.3 (Exterior-sphere normals). For $x \in \partial D$ and $r > 0$, define

$$\mathcal{N}_{x,r} := \left\{ n \in \mathbb{R}^d : |n| = 1, B(x - rn, r) \cap D = \varnothing \right\}, \qquad \mathcal{N}_x := \bigcup_{r>0} \mathcal{N}_{x,r}.$$

A vector in $\mathcal{N}_x$ is an inward unit normal. The domain satisfies a uniform exterior-sphere condition if there is $r_0 > 0$ such that $\mathcal{N}_x = \mathcal{N}_{x,r_0} \neq \varnothing$ for every $x \in \partial D$.

Proposition 2.4 (Equivalent exterior-sphere inequality). *For $x \in \partial D$, $r > 0$, and a unit vector n, the following are equivalent:*

(i) $n \in \mathcal{N}_{x,r}$;

(ii) for every $y \in \overline{D}$,

$$\langle y - x, n \rangle + \frac{|y - x|^2}{2r} \geq 0.$$

Proof. The exterior-ball condition is $|y - (x - rn)|^2 \geq r^2$ for all $y \in \overline{D}$. Expanding the square gives exactly the inequality in (ii). Conversely, the inequality gives the closed ball exclusion. If an interior point lay on the sphere, moving it slightly toward the centre would produce an interior point inside the excluded ball, so the open ball is disjoint from D. □

The exterior-sphere condition alone does not justify every uniqueness claim for a nonconvex domain. A standard additional hypothesis prevents nearby normal directions from becoming tangential to one another.

Definition 2.5 (Uniform non-tangency condition). The domain satisfies a uniform non-tangency condition if there are $\delta > 0$ and $\beta \geq 1$ such that, for every $x \in \partial D$, one can choose a unit vector q_x satisfying

$$\langle q_x, n \rangle \geq \frac{1}{\beta}$$

for every $y \in \partial D \cap B(x, \delta)$ and every $n \in \mathcal{N}_y$.

Bounded C^2 domains satisfy both geometric conditions, as do many convex and piecewise smooth domains after the appropriate formulation at corners. For nonsmooth domains the normal may be set valued, so the direction used by the regulator must be recorded explicitly.

Definition 2.6 (Normal Skorokhod problem)**.** Let $f \in C([0,T];\mathbb{R}^d)$ with $f(0) \in \overline{D}$. A triple (g, ℓ, n) solves the normal Skorokhod problem if

(i) $g \in C([0,T];\overline{D})$;

(ii) ℓ is continuous, nondecreasing, and $\ell(0) = 0$;

(iii) n is measurable with respect to the Stieltjes measure $\mathrm{d}\ell$, $|n_s| = 1$, and $n_s \in \mathcal{N}_{g_s}$ for $\mathrm{d}\ell$-almost every s;

(iv)
$$g_t = f_t + \int_0^t n_s \,\mathrm{d}\ell_s, \qquad \int_0^T \mathbf{1}_{\{g_s \in D\}} \,\mathrm{d}\ell_s = 0.$$

Remark 2.7. For a bounded C^2 domain, the inward normal is single valued and continuous on the boundary. The normal Skorokhod problem is then well posed, and its map is continuous on compact time intervals. In a general nonconvex or nonsmooth domain, existence, uniqueness, and continuity require geometric hypotheses such as the exterior-sphere and non-tangency conditions above; one cannot infer pathwise uniqueness from the exterior-sphere condition alone.

2.2.1 Penalization and its sign

Suppose D is smooth and a C^2 function $p : \mathbb{R}^d \to [0, \infty)$ satisfies $p = 0$ on $\overline{D}$ and, near $\overline{D}$, $p(x) = \mathrm{dist}(x, \overline{D})^2$. Outside the domain, ∇p points outward. Hence a penalized approximation to inward normal reflection has the form

$$g_t^\varepsilon = f_t - \frac{1}{2\varepsilon} \int_0^t \nabla p(g_s^\varepsilon) \,\mathrm{d}s, \tag{2.2}$$

not the same formula with a plus sign. The factor $1/2$ is conventional and matches $\nabla \,\mathrm{dist}^2 = 2 \,\mathrm{dist}\, n_{\mathrm{out}}$ where the nearest-point projection is smooth.

Theorem 2.8 (Reflected SDE in a smooth domain)**.** *Let D be a bounded C^2 domain and let $n(x)$ be its inward unit normal. Consider*

$$X_t = X_0 + \int_0^t a(s, X_s) \,\mathrm{d}s + \int_0^t \sigma(s, X_s) \,\mathrm{d}B_s + \int_0^t n(X_s) \,\mathrm{d}L_s. \tag{2.3}$$

If a and σ are globally Lipschitz in x, uniformly in t, and have linear growth, then (2.3) has a unique strong solution for every square-integrable $X_0 \in \overline{D}$. If instead the coefficients are bounded and continuous and the covariance is uniformly nondegenerate, there is at least a weak solution.

Proof outline. The deterministic normal Skorokhod map on a bounded C^2 domain is nonanticipative and continuous. Euler approximations combined with tightness give weak existence. Under Lipschitz coefficients, the deterministic stability estimate and the usual stochastic estimates give pathwise uniqueness. Weak existence plus pathwise uniqueness then yields a strong solution. □

2.3 General reflection fields

Let $\mathcal{K}(x) \subset \mathbb{R}^d$ be a nonempty closed set of admissible inward unit directions for each $x \in \partial D$. Normalizing directions is essential: without it, the same finite-variation push can be represented by rescaling a direction and the scalar regulator in infinitely many ways.

Definition 2.9 (General Skorokhod problem). Let $f \in C([0,T];\mathbb{R}^d)$ with $f(0) \in \overline{D}$. A pair (g,k) solves the Skorokhod problem for $(D,\mathcal{K},f)$ if

(i) $g \in C([0,T];\overline{D})$ and k is continuous with finite variation and $k_0 = 0$;

(ii) $g_t = f_t + k_t$;

(iii) the variation measure $\mathrm{d}|k|$ is supported on $\{t : g_t \in \partial D\}$;

(iv) there is a $\mathrm{d}|k|$-measurable process γ such that

$$k_t = \int_0^t \gamma_s \,\mathrm{d}|k|_s, \qquad \gamma_s \in \mathcal{K}(g_s), \quad |\gamma_s| = 1, \quad \mathrm{d}|k|\text{-a.e.}$$

If (g,k) is unique, write $\overline{\Gamma}(f) = (\Gamma(f), \mathcal{R}(f)) := (g,k)$ and call $\overline{\Gamma}$ the extended Skorokhod map.

Proposition 2.10 (Nonanticipativity). *If the Skorokhod problem is unique on every finite interval, then the extended map is nonanticipative: if f and $\tilde{f}$ agree on $[0,t]$, then their solutions agree on $[0,t]$.*

Proof. Restrict both solutions to $[0,t]$. They solve the same deterministic problem there, so uniqueness gives equality. Continuity of the map is not needed for this argument. □

Let $a : [0,T] \times \overline{D} \to \mathbb{R}^d$ and $\sigma : [0,T] \times \overline{D} \to \mathbb{R}^{d\times m}$.

Definition 2.11 (Strong reflected SDE). On a prescribed stochastic basis carrying an m-dimensional Brownian motion B, a strong solution is a pair (X,K) of continuous adapted processes such that $X_t \in \overline{D}$, K has finite variation, and

$$X_t = X_0 + \int_0^t a(s,X_s)\,\mathrm{d}s + \int_0^t \sigma(s,X_s)\,\mathrm{d}B_s + K_t, \tag{2.4}$$

where K has a polar decomposition $K_t = \int_0^t \gamma_s \,\mathrm{d}|K|_s$ with $\gamma_s \in \mathcal{K}(X_s)$ and $|\gamma_s| = 1$ for $\mathrm{d}|K|$-almost every s, and $\mathrm{d}|K|$ is supported on $\{s : X_s \in \partial D\}$.

Definition 2.12 (Weak reflected SDE). A weak solution consists of a filtered probability space, an m-dimensional Brownian motion on that space, an initial variable with the prescribed law, and adapted processes (X,K) satisfying all conditions in the preceding definition. The stochastic basis and the Brownian motion are part of the unknown.

Whenever the extended Skorokhod map is defined, (2.4) is equivalently

$$\eta_t = X_0 + \int_0^t a(s,\Gamma(\eta)_s)\,\mathrm{d}s + \int_0^t \sigma(s,\Gamma(\eta)_s)\,\mathrm{d}B_s,$$

$$(X, K) = \overline{\Gamma}(\eta).$$

2.4 Weak existence from a continuous Skorokhod map

Theorem 2.13 (Abstract weak existence)**.** *Assume that the extended Skorokhod map is uniquely defined and continuous for uniform convergence on compact time intervals. Suppose a and σ are bounded and continuous, and let the initial law be supported on $\overline{D}$. Then the reflected SDE (2.4) has a weak solution on every finite horizon.*

Proof. Fix $T > 0$ and a partition with mesh tending to zero. Let $\kappa_n(s)$ be the left endpoint of the cell containing s. Construct the Euler scheme

$$\eta_t^n = X_0^n + \int_0^t a\big(s, X^n_{\kappa_n(s)}\big)\ \mathrm{d}s + \int_0^t \sigma\big(s, X^n_{\kappa_n(s)}\big)\ \mathrm{d}B_s,$$
$$(X^n, K^n) = \overline{\Gamma}(\eta^n).$$

Nonanticipativity makes the recursion well defined.

The drift parts are uniformly Lipschitz in time. For the martingale parts, the Burkholder–Davis–Gundy inequality and boundedness of σ give

$$\mathbb{E}|M_t^n - M_s^n|^4 \le C|t-s|^2.$$

Kolmogorov's tightness criterion therefore makes (η^n) tight in $C([0,T];\mathbb{R}^d)$. Continuity of $\overline{\Gamma}$ makes (η^n, X^n, K^n) jointly tight.

Take a convergent subsequence. The vanishing mesh and continuity of the coefficients imply convergence of the drift characteristics. Rather than claiming convergence of stochastic integrals driven by changing Brownian motions, identify the limit martingale M through its conditional moments:

$$M_t := \eta_t - X_0 - \int_0^t a(s, X_s)\ \mathrm{d}s,$$
$$\langle M\rangle_t = \int_0^t \sigma(s, X_s)\sigma(s, X_s)^\top\ \mathrm{d}s.$$

The discrete martingale identities pass to the limit, so M is a continuous martingale with the displayed bracket. The martingale representation theorem, after enlarging the space if necessary, produces an m-dimensional Brownian motion $\widetilde{B}$ such that $M_t = \int_0^t \sigma(s, X_s)\ \mathrm{d}\widetilde{B}_s$. Finally, continuity of the Skorokhod map gives $(X, K) = \overline{\Gamma}(\eta)$. Thus the limit is a weak solution. □

Remark 2.14 (Why the martingale identification is needed)*.* A Skorokhod representation may place the nth approximation with a different Brownian motion on a new probability space. Uniform convergence of those Brownian motions and their integrands does not, by itself, imply L^2

convergence of the stochastic integrals. The martingale and quadratic-variation identities provide the required stable passage to the limit.

Theorem 2.15 (Continuous dependence)**.** *Let a_n, σ_n be uniformly bounded and converge to a_0, σ_0 locally uniformly on $[0, T] \times \overline{D}$. Let the initial laws μ_n converge weakly to μ_0, and assume that the same continuous extended Skorokhod map is used for every n.*
If the limit reflected SDE has uniqueness in law, then

$$X^n \Longrightarrow X^0 \quad \text{in } C([0, T]; \overline{D}).$$

If, moreover, all equations are driven on one stochastic basis by the same Brownian motion, the initial variables converge in probability, and the limit equation has pathwise uniqueness, then

$$\sup_{0 \le t \le T} |X_t^n - X_t^0| \xrightarrow{\mathbb{P}} 0.$$

Proof. The tightness estimates from the weak-existence proof are uniform in n. Every subsequential limit solves the equation with coefficients (a_0, σ_0), by the same martingale-characteristic argument and the continuity of $\overline{\Gamma}$. Uniqueness in law identifies every limit and therefore proves weak convergence of the full sequence.
For the common-noise statement, apply the same argument to every pair of subsequences. Any joint limit consists of two solutions of the limit equation with the same Brownian motion and the same initial variable. Pathwise uniqueness forces the two limits to coincide. The Gyöngy–Krylov criterion then yields convergence in probability. □

Chapter 3 PDE and Submartingale Characterizations

3.1 Reflected diffusions and parabolic PDEs

Let $D \subset \mathbb{R}^d$ be a domain for which the reflected SDE is well posed in law. In the time-homogeneous case, write

$$X_t = x + \int_0^t a(X_s)\,\mathrm{d}s + \int_0^t \sigma(X_s)\,\mathrm{d}B_s + \int_0^t \gamma(X_s)\,\mathrm{d}L_s, \tag{3.1}$$

where γ is an inward reflection field and $\mathrm{d}L$ is supported on the boundary. Set

$$c(x) := \sigma(x)\sigma(x)^\top$$

and define, for $\varphi \in C^2(\overline{D})$,

$$\mathcal{A}\varphi(x) := a(x) \cdot \nabla\varphi(x) + \frac{1}{2}\operatorname{tr}\Big(c(x)D^2\varphi(x)\Big). \tag{3.2}$$

The oblique derivative at the boundary is $\partial_\gamma\varphi := \nabla\varphi \cdot \gamma$.

Uniqueness in law is needed here: weak existence alone does not select a unique family of transition probabilities. Under well posedness, define

$$P_t\varphi(x) := \mathbb{E}_x[\varphi(X_t)].$$

Proposition 3.1 (A generator core candidate)**.** *Assume a, σ, and γ are bounded and continuous. If $\varphi \in C_b^2(\overline{D})$ satisfies $\partial_\gamma\varphi = 0$ on ∂D, then*

$$\lim_{t\downarrow 0} \frac{P_t\varphi(x) - \varphi(x)}{t} = \mathcal{A}\varphi(x), \qquad x \in \overline{D}. \tag{3.3}$$

Proof. Itô's formula gives

$$\begin{aligned}\varphi(X_t) - \varphi(x) = \int_0^t \mathcal{A}\varphi(X_s)\,\mathrm{d}s + \int_0^t \nabla\varphi(X_s)\sigma(X_s)\,\mathrm{d}B_s \\ + \int_0^t \partial_\gamma\varphi(X_s)\,\mathrm{d}L_s.\end{aligned}$$

The last term is zero because $\mathrm{d}L$ is supported on ∂D, and the stochastic integral is a true martingale because its integrand is bounded. After taking expectations, continuity of X and bounded continuity of $\mathcal{A}\varphi$ give (3.3) by dominated convergence. □

Remark 3.2 (What the proposition does not prove). The proposition identifies a natural class of functions on which the probabilistic generator acts as $\mathcal{A}$ with an oblique Neumann boundary condition. It does not identify the whole generator domain, prove that this class is a core, or prove strong continuity of (P_t). A pointwise derivative at $t = 0$ cannot replace the separate Feller and strong-continuity arguments required by semigroup theory.

3.1.1 The backward Kolmogorov equation

For time-dependent coefficients, let $X^{t,x}$ be the reflected diffusion started from x at time t, and define

$$\begin{aligned} \mathcal{A}_s\varphi(x) &:= a(s,x)\cdot\nabla\varphi(x) + \frac{1}{2}\operatorname{tr}\Big(c(s,x)D^2\varphi(x)\Big), \\ c(s,x) &:= \sigma(s,x)\sigma(s,x)^\top. \end{aligned}$$

The natural probabilistic object is now a transition family $P_{t,s}$, not a one-parameter semigroup.

Theorem 3.3 (Classical backward representation). *Let $u \in C^{1,2}([0,T]\times\overline{D})$ satisfy*

$$\begin{aligned} \partial_t u(t,x) + \mathcal{A}_t u(t,x) &= 0, && (t,x)\in[0,T)\times D, \\ \partial_{\gamma(t,\cdot)}u(t,x) &= 0, && (t,x)\in[0,T)\times\partial D, \\ u(T,x) &= g(x), && x\in\overline{D}. \end{aligned}$$

Assume that u, its derivatives appearing above, and the stochastic integrand $\nabla u\,\sigma$ have enough growth control to justify Itô's formula and the martingale expectation. Then

$$u(t,x) = \mathbb{E}\big[g(X_T^{t,x})\big]. \tag{3.4}$$

Proof. Apply Itô's formula to $u(s, X_s^{t,x})$ on $[t,T]$. The time–space drift vanishes by the PDE, and the finite-variation boundary term vanishes because $\partial_\gamma u = 0$ where the regulator increases. The remaining stochastic integral is a martingale. Taking expectations gives (3.4). □

Remark 3.4. Existence of a classical solution and existence of a transition density are separate analytic questions. On a sufficiently smooth domain, uniformly parabolic operators with compatible Hölder data admit classical Neumann or oblique-derivative theory. When a transition density exists, the function in (3.4) may be written as its integral against g. A PDE solution is a deterministic function; it is not itself a Markov process.

Theorem 3.5 (Feynman–Kac formula). *Let r be bounded from below, and let $u \in C^{1,2}([0,T]\times\overline{D})$ solve*

$$\begin{aligned} \partial_t u + \mathcal{A}_t u - ru &= 0 && \text{on } [0,T)\times D, \\ \partial_\gamma u &= 0 && \text{on } [0,T)\times\partial D, \\ u(T,\cdot) &= g. \end{aligned}$$

Assume the exponential weight and the stochastic integral below are integrable. Then

$$u(t,x) = \mathbb{E}\left[g(X_T^{t,x}) \exp\left(-\int_t^T r(s, X_s^{t,x})\, \mathrm{d}s\right)\right]. \tag{3.5}$$

Proof. Apply the product rule to

$$\exp\left[-\int_t^s r(q, X_q^{t,x})\, \mathrm{d}q\right] u(s, X_s^{t,x}).$$

The PDE cancels the drift and the boundary condition cancels the regulator term. The remaining local martingale is a true martingale under the stated integrability assumption. Evaluating at t and T proves the formula. □

3.1.2 Expected hitting times

Let K be an open set with $\overline{K} \subset D$, put $U := D \setminus \overline{K}$, and define

$$\tau_K := \inf\{t \geq 0 : X_t \in \overline{K}\}.$$

Theorem 3.6 (Poisson problem for the mean hitting time)**.** *Assume D is bounded and that the stopped stochastic integrals below are true martingales. Suppose u extends to a C^2 function on a neighbourhood of $\overline{U}$ and satisfies*

$$\begin{aligned} \mathcal{A}u &= -1 && \text{in } U, \\ \partial_\gamma u &= 0 && \text{on } \partial D, \\ u &= 0 && \text{on } \partial K. \end{aligned}$$

Then $\tau_K < \infty$ almost surely and

$$u(x) = \mathbb{E}_x[\tau_K], \qquad x \in \overline{U}. \tag{3.6}$$

Proof. Itô's formula stopped at $\tau_K \wedge t$ yields

$$\mathbb{E}_x u(X_{\tau_K \wedge t}) = u(x) - \mathbb{E}_x(\tau_K \wedge t).$$

Because u is bounded on $\overline{U}$, the right-hand side gives a uniform bound on $\mathbb{E}_x(\tau_K \wedge t)$. Monotone convergence then implies $\mathbb{E}_x \tau_K < \infty$, and hence $\tau_K < \infty$ almost surely. Continuity of X gives $X_{\tau_K} \in \partial K$; bounded convergence and the inner Dirichlet condition now give (3.6). □

Example 3.7 (Reflected Brownian motion in an annulus)**.** Let $D = B(0, R)$ and $K = B(0, r)$ with $0 < r < R$. Let X be Brownian motion normally reflected at ∂D, so its generator is $\frac{1}{2}\Delta$. The mean time to hit $\overline{K}$ is radial: $u(x) = U(|x|)$. For $r \leq \rho \leq R$, it solves

$$U''(\rho) + \frac{d-1}{\rho} U'(\rho) = -2, \qquad U(r) = 0, \qquad U'(R) = 0.$$

Thus, for $d \geq 3$,

$$u(x) = \frac{r^2 - |x|^2}{d} + \frac{2R^d}{d(d-2)}\left(r^{2-d} - |x|^{2-d}\right), \tag{3.7}$$

where the factor R^d is required by the Neumann condition. For $d = 2$,

$$u(x) = \frac{r^2 - |x|^2}{2} + R^2 \log \frac{|x|}{r}. \tag{3.8}$$

For $d = 1$, on either component of $(-R, R) \setminus [-r, r]$,

$$u(x) = r^2 - |x|^2 + 2R(|x| - r).$$

3.2 The submartingale problem

The submartingale formulation characterizes reflected diffusions without choosing a Brownian motion in advance. Let

$$G = \{x \in \mathbb{R}^d : \phi(x) > 0\}, \qquad \partial G = \{x : \phi(x) = 0\},$$

where $\phi \in C_b^2(\mathbb{R}^d)$ and $|\nabla \phi|$ is bounded away from zero on the boundary. Let $a(t,x)$ and $c(t,x)$ be the interior drift and covariance, let $\gamma(t,x)$ be a continuous inward field with

$$\inf_{t,x \in \partial G} \gamma(t,x) \cdot \nabla \phi(x) > 0,$$

and let $\rho(t,x) \geq 0$ describe possible boundary holding. Define

$$\begin{aligned}
\mathcal{A}_t f(t,x) &:= a(t,x) \cdot \nabla f(t,x) + \frac{1}{2} \operatorname{tr}\left(c(t,x) D^2 f(t,x)\right), \\
\mathcal{J}_t f(t,x) &:= \gamma(t,x) \cdot \nabla f(t,x).
\end{aligned}$$

Definition 3.8 (Submartingale problem)**.** A probability measure P on $C([t_0,\infty);\mathbb{R}^d)$ solves the submartingale problem for (a, c, γ, ρ) with initial point $x_0 \in \overline{G}$ if, for the coordinate process X,

(i)

$$P\left(X_{t_0} = x_0,\ X_t \in \overline{G} \text{ for every } t \geq t_0\right) = 1;$$

(ii) for every compactly supported $f \in C^{1,2}([t_0,\infty) \times \mathbb{R}^d)$ satisfying

$$\rho(t,x)\partial_t f(t,x) + \mathcal{J}_t f(t,x) \geq 0 \quad \text{on } [t_0,\infty) \times \partial G, \tag{3.9}$$

the process

$$f(t, X_t) - f(t_0, X_{t_0}) - \int_{t_0}^{t} \mathbf{1}_{\{X_s \in G\}} \left(\partial_s f + \mathcal{A}_s f\right)(s, X_s)\, \mathrm{d}s \tag{3.10}$$

is a P-submartingale.

The event in (i) is a pathwise confinement statement. Requiring only $P(X_t \in \overline{G}) = 1$ separately for each fixed t would be weaker in form, even though continuity can sometimes be used to recover the pathwise version from a countable dense set of times.

Theorem 3.9 (Semimartingale representation)**.** *Under the usual smoothness and nondegeneracy assumptions for the submartingale problem, P solves the problem above if and only if there is a continuous adapted nondecreasing process L, with $L_{t_0} = 0$ and*

$$\int_{t_0}^{t} \mathbf{1}_{\{X_s \in G\}} \, \mathrm{d}L_s = 0, \tag{3.11}$$

such that

$$M_t := X_t - X_{t_0} - \int_{t_0}^{t} \mathbf{1}_{\{X_s \in G\}} a(s, X_s) \, \mathrm{d}s - \int_{t_0}^{t} \gamma(s, X_s) \, \mathrm{d}L_s \tag{3.12}$$

$$\langle M^i, M^j \rangle_t = \int_{t_0}^{t} \mathbf{1}_{\{X_s \in G\}} c_{ij}(s, X_s) \, \mathrm{d}s, \tag{3.13}$$

$$\int_{t_0}^{t} \mathbf{1}_{\{X_s \in \partial G\}} \, \mathrm{d}s = \int_{t_0}^{t} \rho(s, X_s) \, \mathrm{d}L_s. \tag{3.14}$$

Here M is a continuous local martingale. Under boundedness, it is a true martingale on finite intervals.

Remark 3.10. When $\rho \equiv 0$, relation (3.14) says that the process spends zero Lebesgue time on the boundary. Positive ρ produces sticky or elastic boundary behavior. The process L is part of the representation and must start at zero, increase only on the boundary, and use the bracket lower limit t_0; omitting any of these conditions leaves the characterization incomplete.

Theorem 3.11 (A safe well-posedness regime)**.** *Suppose G is a bounded C^2 domain. Assume that a and a square root σ of c are bounded and globally Lipschitz in x, continuous in t, and that c is uniformly elliptic. Assume γ is bounded, Lipschitz, and uniformly inward pointing.*

For $\rho \equiv 0$, the reflected SDE is pathwise unique and its law is the unique solution of the submartingale problem. Its transition family is strong Markov; equivalently, the time–space process is a homogeneous strong Markov process.

Assume in addition that a, c, γ, and ρ are time homogeneous and that ρ is bounded, nonnegative, and Lipschitz. Let $(\overline{X}, \overline{L})$ denote the nonsticky reflected diffusion and set

$$A_s = s + \int_{t_0}^{s} \rho(\overline{X}_r) \, \mathrm{d}\overline{L}_r, \qquad \tau_t := \inf\{s \geq t_0 : A_s > t\}, \qquad t \geq t_0.$$

Then $X_t := \overline{X}_{\tau_t}$ yields the elastic process, and the corresponding submartingale problem is well posed.

Proof outline. The smooth-domain reflected SDE has a unique strong solution under the stated Lipschitz assumptions. Itô's formula proves that its law solves the submartingale problem. Conversely, the semimartingale representation and the martingale representation theorem recover a weak reflected SDE from any solution of the submartingale problem; uniqueness in law identifies it. In the time-homogeneous elastic case, the inverse clock τ slows physical time exactly when $\overline{X}$ accumulates boundary regulator. The change-of- variables formula then gives (3.14). The construction is reversible because A is continuous and strictly increasing. □

Example 3.12 (Why time-dependent test functions matter)**.** Take $G = (0, \infty)$, $a = 0$, $c = 1$, $\gamma = 1$, and $\rho = 0$. Reflected Brownian motion solves the submartingale problem. Brownian motion stopped upon first hitting zero does not solve it, because after the hitting time a time-dependent test function can change through its boundary trace while the interior integral in (3.10) is switched off. Nevertheless, for every time-independent $f \in C_c^2(\mathbb{R})$ with $f'(0) \geq 0$, the stopped process satisfies

$$f(X_t) - \frac{1}{2}\int_0^t \mathbf{1}_{\{X_s>0\}} f''(X_s)\, \mathrm{d}s$$

as a submartingale (indeed, as a martingale). Therefore one cannot replace the full boundary condition $\rho\, \partial_t f + \mathcal{J} f \geq 0$ by a condition involving only f and its spatial derivatives.

Remark 3.13 (Localization)*.* Boundedness assumptions may be localized. If two sets of coefficients agree on $[t_0, t_0 + \varepsilon] \times U$, where U is a neighborhood of the starting point, then their stopped submartingale problems agree up to the first exit from that time–space region, provided the localized problem is well posed.

Chapter 4 Reflected Backward Stochastic Differential Equations

4.1 Definitions, optimal stopping, and a priori estimates

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \le t \le T}, \mathbb{P})$ be the augmented filtration of a d-dimensional Brownian motion B. We work with scalar reflected BSDEs. Use the spaces

$$\mathbb{S}^2 := \left\{ Y : Y \text{ is continuous and adapted}, \mathbb{E} \sup_{0 \le t \le T} |Y_t|^2 < \infty \right\},$$
$$\mathbb{H}^2 := \left\{ Z : Z \text{ is predictable}, \mathbb{E} \int_0^T |Z_t|^2 \, \mathrm{d}t < \infty \right\},$$
$$\mathbb{A}^2 := \left\{ K : K \text{ is continuous, adapted, and nondecreasing}, K_0 = 0,\ \mathbb{E} K_T^2 < \infty \right\}.$$

The data are a terminal variable ξ, a driver f, and a lower obstacle S. We impose the standing assumptions

(A1) $\xi \in L^2(\mathcal{F}_T)$;

(A2) $f(t, \omega, y, z)$ is progressively measurable in (t, ω) for each (y, z) and there is a constant λ such that

$$|f(t, y, z) - f(t, y', z')| \le \lambda \big(|y - y'| + |z - z'| \big);$$

(A3)

$$\mathbb{E} \left(\int_0^T |f(t, 0, 0)| \, \mathrm{d}t \right)^2 < \infty;$$

(A4) S is continuous and adapted, $\mathbb{E} \sup_{t \le T} (S_t^+)^2 < \infty$, and $S_T \le \xi$ almost surely.

Definition 4.1 (Reflected BSDE)**.** A triple $(Y, Z, K) \in \mathbb{S}^2 \times \mathbb{H}^2 \times \mathbb{A}^2$ solves the reflected BSDE with data (ξ, f, S) if

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) \, \mathrm{d}s - \int_t^T Z_s \, \mathrm{d}B_s + K_T - K_t, \tag{4.1}$$

$$Y_t \ge S_t, \tag{4.2}$$

$$\int_0^T (Y_t - S_t) \, \mathrm{d}K_t = 0. \tag{4.3}$$

The last identity is the Skorokhod minimality condition: K may increase only on the contact set $\{Y = S\}$.

For $t \in [0,T]$, let $\mathcal{T}_t$ be the stopping times with values in $[t,T]$, and set

$$\widehat{S}_\tau := S_\tau \mathbf{1}_{\{\tau<T\}} + \xi \mathbf{1}_{\{\tau=T\}}.$$

Theorem 4.2 (First contact and optimal stopping). *Let (Y,Z,K) solve the reflected BSDE and define*

$$\tau_t := \inf\{s \ge t : Y_s = S_s\} \wedge T.$$

Then

(i) $K_{\tau_t} = K_t$ *and*

$$Y_{\tau_t} = S_{\tau_t} \mathbf{1}_{\{\tau_t<T\}} + \xi \mathbf{1}_{\{\tau_t=T\}};$$

(ii) pathwise,

$$K_T - K_t = \sup_{t\le s\le T} \left(\xi + \int_s^T f(r,Y_r,Z_r)\,\mathrm{d}r - \int_s^T Z_r\,\mathrm{d}B_r - S_s \right)^- ; \tag{4.4}$$

(iii)

$$Y_t = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_t} \mathbb{E}_t\left[\int_t^\tau f(s,Y_s,Z_s)\,\mathrm{d}s + \widehat{S}_\tau \right], \tag{4.5}$$

and τ_t is optimal.

Proof. On $[t,\tau_t)$ one has $Y > S$, so the minimality condition gives $K_s = K_t$. Continuity gives the stated value at τ_t.

For $s \in [t,T]$, the BSDE identity implies

$$\xi + \int_s^T f(r,Y_r,Z_r)\,\mathrm{d}r - \int_s^T Z_r\,\mathrm{d}B_r - S_s = (Y_s - S_s) - (K_T - K_s).$$

Its negative part is at most $K_T - K_s \le K_T - K_t$. At $s = \tau_t < T$, $Y_s = S_s$ and $K_s = K_t$, so equality holds. If $\tau_t = T$, then K is constant on $[t,T]$ and both sides of (4.4) are zero.

For any $\tau \in \mathcal{T}_t$, stop (4.1) at τ. Since $Y_\tau \ge \widehat{S}_\tau$ and $K_\tau - K_t \ge 0$, conditional expectation gives the inequality "$\ge$" in (4.5). At $\tau = \tau_t$, the regulator is constant and the reward equals Y_{τ_t}, so equality holds. $\square$

4.1.1 Comparison with nonreflected equations

Proposition 4.3. *Let (Y, Z, K) solve the reflected BSDE. Let $(\underline{Y}, \underline{Z})$ solve the ordinary BSDE with the same terminal condition and driver,*

$$\underline{Y}_t = \xi + \int_t^T f(s, \underline{Y}_s, \underline{Z}_s)\ \mathrm{d}s - \int_t^T \underline{Z}_s\ \mathrm{d}B_s.$$

Let $\overline{Y}$ solve the forward equation with the fixed process Z,

$$\overline{Y}_t = Y_0 - \int_0^t f(s, \overline{Y}_s, Z_s)\ \mathrm{d}s + \int_0^t Z_s\ \mathrm{d}B_s.$$

Then

$$\underline{Y}_t \leq Y_t \leq \overline{Y}_t \qquad \text{for all } t, \quad \mathbb{P}\text{-a.s.}$$

Proof. The lower inequality is the scalar BSDE comparison theorem: the reflected equation has the additional nonnegative backward increment $K_T - K_t$.
For the upper inequality, set $D_t := Y_t - \overline{Y}_t$. Then $D_0 = 0$ and

$$\mathrm{d}D_t = -\alpha_t D_t\ \mathrm{d}t - \mathrm{d}K_t,$$

where α is a bounded predictable linearization of $y \mapsto f(t, y, Z_t)$. Multiplying by $\exp(\int_0^t \alpha_s\ \mathrm{d}s)$ shows that the transformed difference is nonincreasing, and hence $D_t \leq 0$. □

4.1.2 A priori and stability estimates

Write $Y^* := \sup_{0 \leq t \leq T} |Y_t|$ and define

$$I(\xi, f, S)^2 := \mathbb{E}\left[|\xi|^2 + \left(\int_0^T |f(t, 0, 0)|\ \mathrm{d}t \right)^2 + \sup_{0 \leq t \leq T} (S_t^+)^2 \right]. \tag{4.6}$$

Theorem 4.4 (A priori estimate)**.** *There is a constant $C = C(T, \lambda)$ such that every solution satisfies*

$$\mathbb{E}\left[(Y^*)^2 + \int_0^T |Z_t|^2\ \mathrm{d}t + K_T^2 \right] \leq C I(\xi, f, S)^2. \tag{4.7}$$

Proof. Apply Itô's formula to $e^{\beta t}|Y_t|^2$, with β large enough. The minimality condition gives

$$\int_0^T Y_t\ \mathrm{d}K_t = \int_0^T S_t\ \mathrm{d}K_t \leq \sup_{t \leq T} S_t^+ K_T.$$

The Lipschitz bound on f, Young's inequality, and a sufficiently large β absorb the Y and Z terms.
Next use

$$K_T = Y_0 - \xi - \int_0^T f(t, Y_t, Z_t)\ \mathrm{d}t + \int_0^T Z_t\ \mathrm{d}B_t$$

to estimate K_T^2. Finally, the Burkholder–Davis–Gundy inequality applied to the BSDE controls Y^*. Choosing the Young constants in this order absorbs every occurrence of K_T^2 and yields (4.7); no unmultiplied K_T^2 term remains on the right-hand side. □

For two data sets, use $\Delta Y := Y^1 - Y^2$ and similarly for the other quantities. Define

$$\delta f_t := f^1(t, Y_t^2, Z_t^2) - f^2(t, Y_t^2, Z_t^2), \qquad I_i := I(\xi_i, f^i, S^i).$$

Theorem 4.5 (Stability)**.** *There is $C = C(T, \lambda)$ such that*

$$\begin{aligned}
&\mathbb{E}\left[\sup_{t\le T} |\Delta Y_t|^2 + \int_0^T |\Delta Z_t|^2 \,\mathrm{d}t + \sup_{t\le T} |\Delta K_t|^2\right] \\
&\quad \le C\mathbb{E}\left[|\Delta\xi|^2 + \left(\int_0^T |\delta f_t| \,\mathrm{d}t\right)^2\right] + C(I_1 + I_2)\left(\mathbb{E}\sup_{t\le T} |\Delta S_t|^2\right)^{1/2}.
\end{aligned} \tag{4.8}$$

In particular, a reflected BSDE with fixed data has at most one solution. The estimate controls the uniform norm of ΔK, not the total variation of $K^1 - K^2$.

Proof. Linearize the difference of the drivers in (y, z) and apply Itô's formula to $e^{\beta t}|\Delta Y_t|^2$. The reflection term has the key bound

$$\begin{aligned}
\int_0^T \Delta Y_t \,\mathrm{d}\Delta K_t &\le \int_0^T \Delta S_t \,\mathrm{d}K_t^1 - \int_0^T \Delta S_t \,\mathrm{d}K_t^2 \\
&\le \sup_{t\le T} |\Delta S_t| (K_T^1 + K_T^2).
\end{aligned}$$

Indeed, $Y^1 = S^1$ on the support of $\mathrm{d}K^1$, while $Y^2 \ge S^2$, and the analogous statement holds for K^2. Young's inequality, the a priori estimate, Gronwall's lemma, and the Burkholder–Davis–Gundy inequality give the Y and Z bounds. Finally, subtracting the two BSDEs expresses ΔK in terms of $\Delta Y, \Delta Z, \Delta\xi$, and δf, which gives the last term in (4.8). □

Theorem 4.6 (Comparison)**.** *For $i = 1, 2$, let (Y^i, Z^i, K^i) solve the scalar reflected BSDE with data (ξ_i, f^i, S^i). Assume*

$$\xi_1 \le \xi_2, \qquad S_t^1 \le S_t^2, \qquad f^1(t, y, z) \le f^2(t, y, z)$$

for all (t, y, z), almost surely. Then $Y_t^1 \le Y_t^2$ for every t, almost surely.

Proof. Apply the Itô–Tanaka formula to $e^{\beta t}(\Delta Y_t^+)^2$. On the set $\{\Delta Y_t > 0\}$ one has $Y_t^1 > Y_t^2 \ge S_t^2 \ge S_t^1$, so $\mathrm{d}K_t^1 = 0$. The remaining reflection term has the favorable sign because $\mathrm{d}K^2 \ge 0$. Linearizing the driver, using its order, and taking β large enough give $\mathbb{E}(\Delta Y_t^+)^2 = 0$. This argument works from every time t; checking only a possible failure at time zero would not prove the comparison theorem. □

4.2 Essential suprema and the Snell envelope

4.2.1 Essential suprema

Definition 4.7. For a family $\mathcal{X} = \{X_i : i \in I\} \subset L^0(\mathcal{F})$, an essential supremum is a random variable X such that

(i) $X \geq X_i$ almost surely for every i;

(ii) every other almost-sure upper bound U satisfies $U \geq X$ almost surely.

It is denoted by $\operatorname{ess\,sup}_{i\in I} X_i$ and is unique up to almost-sure equality. Essential infima are defined by changing signs.

Theorem 4.8 (Countable realization)**.** *Assume $\mathcal{X}$ has an almost-surely finite essential upper bound. Then its essential supremum exists and there are indices (i_n) such that*

$$\operatorname*{ess\,sup}_{i\in I} X_i = \sup_{n\geq 1} X_{i_n} \quad \mathbb{P}\text{-a.s.}$$

If $\mathcal{X}$ is upward directed—for every i, j there is k with $X_k \geq X_i \vee X_j$—the sequence can be chosen increasing.

Proof. Let $\mathcal{C}$ be the set of suprema of countable subfamilies of $\mathcal{X}$, and let $\psi(x) = \arctan x + \pi/2$, a bounded strictly increasing function. Set

$$\alpha := \sup_{Z\in\mathcal{C}} \mathbb{E}[\psi(Z)].$$

Choose $Z_n \in \mathcal{C}$ with $\mathbb{E}\psi(Z_n) \geq \alpha - 1/n$, and put $Z^* := \sup_n Z_n$. The union of the countable subfamilies defining the Z_n is countable, so $Z^* \in \mathcal{C}$. For any i, the variable $Z^* \vee X_i$ also belongs to $\mathcal{C}$. Hence

$$\mathbb{E}\psi(Z^* \vee X_i) \leq \alpha = \mathbb{E}\psi(Z^*).$$

Strict monotonicity of ψ gives $X_i \leq Z^*$ almost surely. Thus Z^* is the least essential upper bound. Under upward directedness, recursively choose one index dominating the first n members of the countable family. This produces an increasing realizing sequence. The argument avoids the circular use of expectations of unbounded Zorn-chain elements. □

4.2.2 The Snell envelope

Let

$$\widehat{S}_t := S_t \mathbf{1}_{\{t<T\}} + \xi \mathbf{1}_{\{t=T\}}, \qquad Y_t := \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_t} \mathbb{E}_t[\widehat{S}_\tau].$$

Theorem 4.9 (Snell envelope). *The process Y has a continuous version in $\mathbb{S}^2$. It is the smallest supermartingale dominating $\widehat{S}$. For*

$$\tau_t^* := \inf\{s \geq t : Y_s = \widehat{S}_s\} \wedge T,$$

one has

$$Y_t = \mathbb{E}_t[\widehat{S}_{\tau_t^*}], \qquad (Y_{s\wedge\tau_t^*})_{s\geq t} \text{ is a martingale.}$$

More generally, for every stopping time $\theta \in \mathcal{T}_t$,

$$Y_t = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_t} \mathbb{E}_t\left[\widehat{S}_\tau \mathbf{1}_{\{\tau<\theta\}} + Y_\theta \mathbf{1}_{\{\tau\geq\theta\}}\right]. \tag{4.9}$$

Proof. For fixed t, the conditional rewards are upward directed: paste two stopping times along the $\mathcal{F}_t$-event on which the first conditional reward is larger. The preceding theorem therefore gives a sequence whose conditional expectations increase to Y_t. Pasting these maximizing sequences at a deterministic or stopping time proves the dynamic programming identity (4.9). Taking $\tau = \theta$ shows that Y is a supermartingale, and the same identity proves its minimality among supermartingales dominating the reward.

The terminal reward may differ from S_T. Put $M_t := \mathbb{E}_t\xi$ and replace the reward, without changing its Snell envelope, by the continuous process

$$\widetilde{S}_t := S_t \vee \big(M_t - (T-t)\big), \qquad \widetilde{S}_T = \xi.$$

Indeed, whenever the second term is selected at a stopping time, waiting until T gives a conditional reward at least as large. The Brownian filtration is continuous, so M is continuous. Dyadic-time Snell envelopes of $\widetilde{S}$ converge in $\mathbb{S}^2$ by Doob's L^2 maximal inequality; the limit is a continuous version of Y.

Apply (4.9) to the stopping times $\tau_n := \inf\{s \geq t : Y_s - \widehat{S}_s \leq 1/n\} \wedge T$. Continuity gives $\tau_n \uparrow \tau_t^*$, while the DPP shows that Y stopped at τ_n is asymptotically a martingale. Uniform integrability and continuity yield $Y_t = \mathbb{E}_t\widehat{S}_{\tau_t^*}$ and the martingale property up to first contact. □

4.3 Well posedness of reflected BSDEs

4.3.1 Frozen drivers and Picard iteration

Proposition 4.10 (Frozen driver). *Let h be progressively measurable and satisfy*

$$\mathbb{E}\left(\int_0^T |h_t| \,\mathrm{d}t\right)^2 < \infty.$$

Then the reflected BSDE with driver h_t has a unique solution.

Proof. Define

$$\widetilde{\xi} := \xi + \int_0^T h_s \, \mathrm{d}s, \qquad \widetilde{S}_t := S_t + \int_0^t h_s \, \mathrm{d}s \quad (t < T),$$

with terminal value $\widetilde{\xi}$. Let $\widetilde{Y}$ be the Snell envelope of this transformed reward. Its Doob–Meyer decomposition in the Brownian filtration is

$$\widetilde{Y}_t = \widetilde{Y}_0 + \int_0^t Z_s \, \mathrm{d}B_s - K_t$$

where $K \in \mathbb{A}^2$. Minimality of the Snell envelope, or equivalently its martingale property before first contact, implies that $\mathrm{d}K$ is supported on $\{\widetilde{Y} = \widetilde{S}\}$. Set $Y_t := \widetilde{Y}_t - \int_0^t h_s \, \mathrm{d}s$. Then (Y, Z, K) solves the desired reflected BSDE. Uniqueness follows from the stability estimate with identical data. □

Theorem 4.11 (Existence and uniqueness). *Under assumptions (A1)–(A4), the reflected BSDE* (4.1)–(4.3) *has a unique solution in* $\mathbb{S}^2 \times \mathbb{H}^2 \times \mathbb{A}^2$.

Proof. For an input pair (U, V), solve the frozen-driver reflected BSDE with $h_t = f(t, U_t, V_t)$, and denote its first two components by $\Phi(U, V) = (Y, Z)$. For two inputs, the obstacles and terminal variables are the same. Consequently,

$$\int_0^T (Y_t - Y'_t) \, \mathrm{d}(K_t - K'_t) \leq 0.$$

Applying Itô's formula to $e^{\beta t}|Y_t - Y'_t|^2$ gives

$$\begin{aligned} \mathbb{E} \int_0^T e^{\beta t} \left(\frac{\beta}{2} |Y_t - Y'_t|^2 + |Z_t - Z'_t|^2 \right) \mathrm{d}t \\ \leq \frac{C\lambda^2}{\beta} \mathbb{E} \int_0^T e^{\beta t} \left(|U_t - U'_t|^2 + |V_t - V'_t|^2 \right) \mathrm{d}t. \end{aligned}$$

For sufficiently large β, Φ is a contraction in the corresponding weighted Hilbert norm. Its fixed point supplies (Y, Z), and K is then defined by the BSDE identity. The frozen-driver construction and convergence show that $K \in \mathbb{A}^2$ and that the minimality condition passes to the limit. Uniqueness also follows from (4.8). □

4.3.2 Penalization

Theorem 4.12 (Penalization approximation). *For* $n \geq 1$, *let* (Y^n, Z^n) *solve*

$$Y_t^n = \xi + \int_t^T \left[f(s, Y_s^n, Z_s^n) + n(S_s - Y_s^n)^+ \right] \mathrm{d}s - \int_t^T Z_s^n \, \mathrm{d}B_s, \tag{4.10}$$

and define

$$K_t^n := n \int_0^t (S_s - Y_s^n)^+ \, \mathrm{d}s.$$

Then Y^n is increasing in n, and for the reflected solution (Y, Z, K),

$$\mathbb{E}\left[\sup_{t\le T}|Y_t^n - Y_t|^2 + \int_0^T |Z_t^n - Z_t|^2 \,\mathrm{d}t + \sup_{t\le T}|K_t^n - K_t|^2\right] \longrightarrow 0. \tag{4.11}$$

Moreover,

$$\mathbb{E}\sup_{t\le T}\big((Y_t^n - S_t)^-\big)^2 \longrightarrow 0.$$

Proof. The drivers in (4.10) increase with n, so ordinary BSDE comparison gives $Y^n \le Y^{n+1}$. Itô's formula, using $Y_t^n \,\mathrm{d}K_t^n \le S_t \,\mathrm{d}K_t^n$, gives the uniform estimate

$$\sup_n \mathbb{E}\left[\sup_{t\le T}|Y_t^n|^2 + \int_0^T |Z_t^n|^2 \,\mathrm{d}t + (K_T^n)^2\right] < \infty.$$

The monotone stability theorem for Lipschitz BSDEs then produces a limit (Y, Z, K) and gives convergence in the norms displayed in (4.11). In particular, the limit Y is continuous. Since

$$\mathbb{E}\int_0^T (S_t - Y_t^n)^+ \,\mathrm{d}t = \frac{1}{n}\mathbb{E}K_T^n \longrightarrow 0,$$

one has $Y \ge S$ for Lebesgue-almost every time and probability-almost every path; continuity of Y and S upgrades this to every time.

For completeness, define the continuous obstacle $S_t^n := Y_t^n \wedge S_t$. Then $Y^n \ge S^n$ and

$$\int_0^T (Y_t^n - S_t^n) \,\mathrm{d}K_t^n = 0.$$

Monotone stability also yields $\mathbb{E}\sup_t |S_t^n - S_t|^2 \to 0$. Passing to the limit in this exact minimality identity proves $\int_0^T (Y_t - S_t) \,\mathrm{d}K_t = 0$. Thus the limit is the reflected solution, and uniqueness identifies the whole sequence. □

4.4 Markovian reflected BSDEs and obstacle PDEs

Let the state dimension be q and consider, for $(t, x) \in [0, T] \times \mathbb{R}^q$,

$$X_s^{t,x} = x + \int_t^s b(r, X_r^{t,x}) \,\mathrm{d}r + \int_t^s \sigma(r, X_r^{t,x}) \,\mathrm{d}B_r, \qquad s \in [t, T]. \tag{4.12}$$

Assume that b and σ are globally Lipschitz in x, have linear growth, and are $1/2$-Hölder in time. Let $f(t, x, y, z)$ be deterministic, Lipschitz in (x, y, z) with linear growth, let g be Lipschitz with linear growth, and let $l \in C^{1,2}([0, T] \times \mathbb{R}^q)$ have bounded derivatives and satisfy $l(T, x) \le g(x)$.

For each (t, x), let $(Y^{t,x}, Z^{t,x}, K^{t,x})$ solve

$$Y_s^{t,x} = g(X_T^{t,x}) + \int_s^T f(r, X_r^{t,x}, Y_r^{t,x}, Z_r^{t,x}) \,\mathrm{d}r - \int_s^T Z_r^{t,x} \,\mathrm{d}B_r + K_T^{t,x} - K_s^{t,x}, \tag{4.13}$$

$$Y_s^{t,x} \ge l(s, X_s^{t,x}), \tag{4.14}$$

$$\int_t^T \left(Y_s^{t,x} - l(s, X_s^{t,x})\right) \, \mathrm{d}K_s^{t,x} = 0. \tag{4.15}$$

Define the decoupling field

$$u(t,x) := Y_t^{t,x}.$$

It is deterministic because it is measurable with respect to the past at time t and, by construction, depends only on Brownian increments after t, which are independent of that past.

Define

$$\mathcal{A}\varphi(t,x) := b(t,x) \cdot D\varphi(t,x) + \frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(t,x) D^2\varphi(t,x)\right), \tag{4.16}$$

$$\mathcal{H}\varphi(t,x) := \partial_t\varphi(t,x) + \mathcal{A}\varphi(t,x) \tag{4.17}$$

$$+ f\left(t, x, \varphi(t,x), D\varphi(t,x)\sigma(t,x)\right). \tag{4.18}$$

The associated obstacle problem is

$$\min\left(u - l, -\mathcal{H}u\right) = 0, \qquad u(T,x) = g(x). \tag{4.19}$$

Equivalently, $\max(l - u, \mathcal{H}u) = 0$.

Theorem 4.13 (Classical verification). *If $u \in C^{1,2}$ is a classical solution of (4.19), then*

$$\begin{aligned} Y_s &:= u(s, X_s^{t,x}), \\ Z_s &:= Du(s, X_s^{t,x})\sigma(s, X_s^{t,x}), \\ K_s &:= -\int_t^s \mathcal{H}u(r, X_r^{t,x}) \, \mathrm{d}r \end{aligned}$$

solve the Markovian reflected BSDE on $[t, T]$.

Proof. The PDE implies $u \geq l$, $\mathcal{H}u \leq 0$, and $\mathcal{H}u = 0$ wherever $u > l$. Hence K is nondecreasing and its measure is supported on the contact set. Itô's formula gives the BSDE, including the stated Z, and the support property gives the Skorokhod condition. □

Theorem 4.14 (Markov property and a coarse modulus). *For $t \leq s \leq T$,*

$$Y_s^{t,x} = u(s, X_s^{t,x}) \quad \mathbb{P}\text{-a.s.} \tag{4.20}$$

Moreover, for a constant C depending only on the stated data,

$$|u(t_1, x_1) - u(t_2, x_2)| \leq C\left(1 + |x_1| + |x_2|\right)^{1/2}$$

$$\times \left(|t_1 - t_2|^{1/4} + |x_1 - x_2|^{1/2}\right). \tag{4.21}$$

Proof. The stochastic flow property for (4.12) and uniqueness of the reflected BSDE give (4.20). Standard forward estimates and (4.8) give

$$|u(t, x_1) - u(t, x_2)|^2 \leq C(1 + |x_1| + |x_2|)|x_1 - x_2|.$$

To estimate time, use the dynamic programming identity over $[t_1, t_2]$:

$$u(t_1, x) = \mathbb{E}\left[u(t_2, X_{t_2}^{t_1,x}) + \int_{t_1}^{t_2} f(s, X_s, Y_s, Z_s)\, \mathrm{d}s + K_{t_2} - K_{t_1}\right].$$

The forward estimate gives $\mathbb{E}|X_{t_2}^{t_1,x} - x|^2 \leq C(1 + |x|^2)(t_2 - t_1)$, so the spatial modulus contributes order $(t_2 - t_1)^{1/4}$. Cauchy–Schwarz and the $\mathbb{H}^2$ estimate give

$$\mathbb{E}\int_{t_1}^{t_2} |f(s, X_s, Y_s, Z_s)|\, \mathrm{d}s \leq C(1 + |x|)(t_2 - t_1)^{1/2}.$$

Under the smooth-obstacle assumptions, the regulator increment has the same or smaller order by the estimate below. Combining the bounds proves (4.21). □

Proposition 4.15 (Lewy–Stampacchia bound for the regulator). *Under the smooth-obstacle assumptions, $K^{t,x}$ is absolutely continuous and*

$$0 \leq \mathrm{d}K_s^{t,x} \leq \mathbf{1}_{\{Y_s^{t,x} = l(s, X_s^{t,x})\}} \kappa_s^{t,x}\, \mathrm{d}s, \tag{4.22}$$

where

$$\kappa_s^{t,x} := \Big[\partial_s l(s, X_s^{t,x}) + \mathcal{A}l(s, X_s^{t,x}) + f\big(s, X_s^{t,x}, l(s, X_s^{t,x}), Dl(s, X_s^{t,x})\sigma(s, X_s^{t,x})\big)\Big]^{-}.$$

Proof outline. Apply the Itô–Tanaka formula to the nonnegative process $Y_s^{t,x} - l(s, X_s^{t,x})$ and use the fact that its negative part vanishes. On the contact set, the martingale coefficient must agree with $Dl\,\sigma$ in the occupation-density sense. The finite-variation part then gives (4.22). The density is the negative part of the obstacle operator evaluated along the contact set. □

4.4.1 Viscosity solutions

Definition 4.16 (Parabolic test functions). Let u be continuous. A function $\varphi \in C^{1,2}$ touches u from above at (t, x) if $u - \varphi$ has a local maximum equal to zero there. It touches u from below if $u - \varphi$ has a local minimum equal to zero there.

Definition 4.17 (Viscosity solution of the obstacle problem). A continuous function u with $u(T, \cdot) = g$ is

(i) a viscosity subsolution if, whenever φ touches u from above at (t, x) with $t < T$ and $u(t, x) > l(t, x)$,

$$\mathcal{H}\varphi(t, x) \geq 0;$$

(ii) a viscosity supersolution if $u \geq l$ and, whenever φ touches u from below at (t, x) with

$t < T$,

$$\mathcal{H}\varphi(t, x) \le 0.$$

It is a viscosity solution if it is both. These are exactly the sub- and supersolution inequalities for the form $\min(u - l, -\mathcal{H}u) = 0$.

Theorem 4.18 (Probabilistic viscosity solution). *The decoupling field $u(t, x) = Y_t^{t,x}$ is a continuous viscosity solution of* (4.19).

Proof. Continuity follows from (4.21); the terminal condition and $u \ge l$ follow directly from the reflected BSDE.

For the subsolution property, let φ touch u from above at (t, x) with $u(t, x) > l(t, x)$. Suppose, to obtain a contradiction, that $\mathcal{H}\varphi(t, x) < 0$. After the standard strictification of the test function, choose a small time–space cylinder on which $u > l$ and $\mathcal{H}\varphi \le -\varepsilon$. Let $\tau > t$ be the first exit of $(s, X_s^{t,x})$ from this cylinder. Before τ the regulator is constant. Put

$$\Delta Y_s := Y_s^{t,x} - \varphi(s, X_s^{t,x}), \qquad \Delta Z_s := Z_s^{t,x} - D\varphi(s, X_s^{t,x})\sigma(s, X_s^{t,x}).$$

Linearizing f and applying an integrating factor (and, equivalently, a bounded Girsanov change of measure) gives

$$\Delta Y_t = \mathbb{E}_t^{\mathbb{Q}}\left[R_\tau \Delta Y_\tau + \int_t^\tau R_s \mathcal{H}\varphi(s, X_s^{t,x})\, \mathrm{d}s\right] < 0,$$

because $\Delta Y_\tau \le 0$ and the integral is strictly negative. This contradicts $\Delta Y_t = u(t, x) - \varphi(t, x) = 0$.

For the supersolution property, let φ touch from below and suppose $\mathcal{H}\varphi(t, x) > 0$. On a small cylinder, $\mathcal{H}\varphi \ge \varepsilon$. The same calculation now includes the nonnegative term $\int_t^\tau R_s\, \mathrm{d}K_s^{t,x}$ and has $\Delta Y_\tau \ge 0$. It therefore yields $\Delta Y_t > 0$, again contradicting contact at (t, x). Hence $\mathcal{H}\varphi(t, x) \le 0$. □